

AMEET YADAV

Senior Backend Developer

ameet.ny1@gmail.com | +91-7385416696 | <https://www.linkedin.com/in/ameet-yadav-a9891726/>

PROFESSIONAL SUMMARY

Senior Backend Software Engineer with 13+ years of experience designing and delivering scalable microservices, event-driven systems, and cloud-native enterprise applications. Hands-on experience building Generative AI and LLM-powered backend services, including prompt engineering, structured model outputs, Retrieval-Augmented Generation (RAG), agentic AI workflows, and integrations using LangChain, LangGraph, Anthropic Claude, and Amazon Bedrock. Strong expertise in Python, Flask, Java, Spring Boot, Apache Kafka, distributed data stores, REST APIs, Docker, Kubernetes, and AWS. Experienced in developing tenant-isolated and resilient services with caching, rate limiting, retries, circuit breakers, observability, and secure AI integration patterns. Proven track record of owning 50-80 production APIs, improving system performance, resolving security vulnerabilities, and mentoring engineering teams.

CORE TECHNICAL SKILLS

Generative AI & LLM Engineering

- Prompt Engineering, Structured Model Outputs, Retrieval-Augmented Generation (RAG)
- Agentic AI Workflows, LangChain, LangGraph, LangSmith, FAISS, chromadb, Pydantic
- Anthropic Claude, AWS Bedrock, Model Routing (Sonnet-class, Haiku-class)
- LangGraph Human-in-the-Loop (interrupt/resume), SQLite Checkpointing, Tool Calling, Groq, Google Gemini Embeddings, Tavily Search, Streamlit

Languages & Frameworks

- Python (Flask), Java (Spring Boot, Spring Cloud, Spring Data JPA), REST APIs

Message Queues & Event Streaming

- Apache Kafka (producers, consumers, topics, partitions, replication), event-driven architecture design, Pub/Sub messaging patterns

Databases & Data Stores

- FAISS, chromadb, PostgreSQL, MySQL, Oracle, Cassandra (NoSQL), Redis, Memcached, database optimization, query tuning, indexing strategies

Cloud & Infrastructure

- AWS (EC2, ECS, EKS, RDS, S3, Secrets Manager, CloudWatch, IAM), Kubernetes (Helm, ConfigMaps, Secrets, StatefulSets), Docker, containerization best practices

DevOps & CI/CD

- GitHub Actions, GitLab CI/CD, monitoring, logging, and observability

Development Tools

- Git, GitHub, GitLab, AI-Assisted Development (GitHub Copilot, Claude), JIRA, Confluence, Agile methodologies

PROFESSIONAL EXPERIENCE

POC Project - Agentic AI Chatbot with LangGraph, RAG & Human-in-the-Loop Approval

Tech Stack: Python, LangGraph, LangChain, Groq (gpt-oss-120b), Google Gemini Embeddings, FAISS, Tavily, SQLite, Streamlit, AWS, CI/CD

- Designed and built an agentic AI chatbot using LangGraph, modelling the agent as a state graph with an LLM node, a tool node, and conditional routing so the model decides when to answer directly or call a tool
- Integrated a Groq-hosted open-weight LLM (gpt-oss-120b) via LangChain tool calling, with a system prompt defining tool-selection rules and guarding against invented answers from uploaded documents
- Built six agent tools: Tavily web search, PDF Retrieval-Augmented Generation (RAG), a restricted-scope calculator, Alpha Vantage stock quotes, OpenWeather current conditions, and a simulated stock purchase
- Implemented the RAG pipeline with PyPDFLoader, recursive text chunking (1000 characters, 200 overlap), Google Gemini embeddings, and a FAISS vector store, returning top-4 similarity matches with source and page metadata
- Implemented Human-in-the-Loop approval using LangGraph interrupt(): the graph pauses before a stock purchase executes, the user approves or rejects in the UI, and execution resumes through Command(resume=...)
- Made pending approvals durable by restoring interrupted state from the checkpoint after page refresh or conversation switching
- Added persistent multi-conversation memory using the LangGraph SQLite checkpointer, with per-thread history and LLM-generated conversation titles stored in SQLite
- Built a Streamlit frontend with token-by-token response streaming, live tool-execution status, in-chat PDF upload, and a conversation sidebar
- Externalized API keys through environment variables and added request timeouts and structured error handling for all third-party API calls
- Deployed the application on AWS through an automated CI/CD pipeline
- Source Code: <https://github.com/AmeetNY/Agentic-Chatbot-using-LanGraph>

Recent Project - Trellix Wise-Style GenAI-Assisted EDR Investigation

- Designed and implemented a Python Flask-based GenAI service for analyzing realistic EDR threat telemetry, generating structured incident summaries, mapping suspicious behavior, and recommending analyst-validated response actions
- Integrated Anthropic Claude through Amazon Bedrock with environment-based configuration and model routing, using a Sonnet-class model for detailed threat analysis and a Haiku-class model for natural-language security search
- Developed mock adapters for Trellix-specific EDR telemetry and tenant provisioning, enabling secure local development and testing without proprietary APIs or customer data
- Enforced tenant isolation across entitlement validation, telemetry access, quotas, rate limits, and cache partitions, with correlation IDs supporting end-to-end request tracing
- Implemented Redis-based rate limiting and daily quotas with an in-memory fallback, plus a DynamoDB-style response cache with TTL-based local fallback
- Improved service resilience using retry and exponential-backoff policies, circuit-breaker behavior, structured error handling, audit logging, service metrics, and cache-hit tracking

- Created structured security prompts and JSON response contracts with prompt-injection protections that treat telemetry as untrusted evidence and require human review before response actions

Microservices & Backend Development

- Architected and developed Python Flask microservices powering enterprise cybersecurity SaaS products serving ~100 enterprise tenants
- Owned backend development for Onboarding Service, Audit Log Exporter, Managed Persistence Service, and Cassandra Service
- Designed and maintained 50-80 production REST APIs with high availability and fault tolerance
- Implemented comprehensive error handling, logging, and monitoring across all services

Event-Driven Architecture & Kafka Integration

- Designed and implemented Kafka-based audit trail using event-driven architecture for real-time event processing
- Built Kafka producers and consumers for asynchronous message processing
- Optimized Kafka topic partitioning and replication strategies for high throughput
- Implemented dead-letter queues and retry mechanisms for failed events

Data Persistence & Optimization

- Designed Cassandra integration for distributed, scalable data storage
- Implemented Memcached integration for caching frequently accessed data, reducing database load by 40%
- Optimized PostgreSQL queries and database schemas for performance
- Managed database migrations and schema versioning

Infrastructure & Deployment

- Deployed and managed applications on Kubernetes using Helm charts, ConfigMaps, and Secrets
- Implemented CI/CD pipelines using GitHub Actions for automated testing and deployment
- Configured ArgoCD for GitOps-based continuous deployment
- Managed application scaling, resource allocation, and pod autoscaling

Feature Management & Security

- Designed in-house feature flag framework replacing LaunchDarkly, enabling granular feature control
- Resolved critical vulnerabilities identified by BlackDuck, Wiz, and Qwiet AI
- Implemented security best practices: secrets management, encryption, and access controls

Code Quality & Mentorship

- Mentored 4-5 interns and 2 QA engineers on backend development best practices
- Conducted comprehensive code reviews and architecture reviews
- Led Threat Investigation Services decommissioning affecting 150+ source files with zero downtime
- Established coding standards and documentation practices

Senior Consultant | CoreCompete (Formerly Accenture)

Mar 2020 - Dec 2021

Microservices Architecture

- Designed and developed cloud-native backend microservices using Java and Spring Boot
- Built scalable REST APIs for workflow orchestration and business process automation

- Implemented service-to-service communication patterns and API versioning strategies

Event-Driven Systems

- Developed event-driven architecture using Google Cloud Pub/Sub for asynchronous message processing
- Built backend services that trigger Apache Airflow DAGs based on Pub/Sub events
- Implemented event serialization, deserialization, and schema management

Containerization & Orchestration

- Deployed applications to Google Kubernetes Engine (GKE)
- Configured resource limits, health checks, and auto-scaling policies
- Implemented CI/CD pipelines for automated deployments

Senior System Analyst | National Stock Exchange of India

Dec 2018 - Feb 2020

Microservices Modernization

- Designed scalable microservice architecture for Trading Platform Modernization
- Implemented API Gateway for request routing and centralized authentication
- Designed service discovery infrastructure

Backend API Development

- Developed backend REST APIs supporting financial transaction processing
- Implemented Spring Cloud architecture for highly available backend services
- Optimized application performance and reliability for high-frequency trading

Distributed Systems

- Improved communication between distributed microservices
- Implemented circuit breakers, load balancing, and failover mechanisms
- Collaborated with multiple development teams to modernize monolithic applications

Associate Consultant | Cognizant Tech Solutions

Jul 2018 - Dec 2018

Kafka Implementation (Banking Domain - Barclays)

- Designed and implemented real-time secure messaging service using Apache Kafka
- Built Kafka producers and consumers for personal and business customer messaging
- Implemented message encryption, authentication, and audit logging
- Ensured compliance with banking security standards and regulations

Senior Software Engineer | Syntel

Apr 2017 - Jun 2018

Backend Services Development

- Designed and developed backend services for Financial Instrument Classification Platform (CFI and FISN codes)
- Developed REST APIs using Spring Boot with comprehensive validation and error handling
- Implemented persistence layer using Spring Data JPA with optimized queries

Software Developer | Saba

Mar 2015 - Dec 2016

Enterprise Backend Development

- Developed enterprise backend applications using Java and Spring Framework
- Designed database access layers using Spring JDBC
- Developed and optimized SQL queries and stored procedures
- Implemented REST services supporting enterprise learning management products
- Performed performance optimization and defect resolution

Junior Software Developer | Saba

Sep 2013 - Feb 2015

- Resolved production defects involving Java memory management and performance issues
- Optimized SQL queries to improve application performance
- Participated in code reviews and software maintenance activities

EDUCATION

Bachelor of Engineering, Computer Science

University of Pune | G.S. Moze College of Engineering, Pune